

Information Lattice picture for the Fisher-Information indicators

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1. Introduction

Here is a rough tl;dr about what this document is about. As can be read in [1] Arnau et. al. had developed a possible interesting way to use the fisher information as a tool to characterise the scrambling-purifying transitions especially in the context of monitored quantum circuits. The very rough idea of how that works is as follows;

It is well known that both product states and maximally entangled states has exponentially decaying correlations. The construction of Arnau et. al. specifically exploits this fact that, the for any such exponentially decaying correlated state, under finite-size scaling the fisher information of well chosen operators stays finite, the hope here is, normally around a critical state, we have scale invariance and hence algebraically correlated states, and hence the fisher information of well chosen operators should diverge under finite-size scaling. This is kind of smart in a sense, especially in context of monitored quantum circuits, where the transition is otherwise hard to characterise. But the catch here is for this to work, one needs to ensure that we are probing correlations that are not exponentially decaying.

NOTE 1 (Questions:). 1. *Is this way of probing things post-selection free?*
 2. *How to choose the operators?*
 3. *Can we actually show that the maximal divergence of fisher information is achieved exactly at the critical point, that is for a conformal state?*

Anyhow, with some further applications of this measure in context of monitored random circuits the method ran into a problem, and the resolution seemed to be that a Random unitary brickwall always generated maximally entangled states, and although local projections disentangles the qubits, on which the measurements were carried on, the rest of the system still remained maximally entangled, and hence the fisher information of any local operator always remained finite. This was kind of a bummer, and one possible way out seems like to use scramblers that generates algebraically correlated states perforated with local projections. The bet is that, in such a scenario, the fisher information of well chosen operators should diverge at the critical point, as it is specifically designed to do so.

On a systematic level, it is clear that the fisher information is designed to probe correlations of states, that falls exactly in the regime of *entanglement barrier*. And it seemed like the most natural choice to decode the physics behind this was to use the *information lattice* by Artiacco et. al. [2], which is specifically designed to probe the information content and entanglement structure of a quantum many-body state at various length scales.

Information lattice is one of the most elegant way to prove scale invariance and algebraic correlations of many-body states. The idea behing the document is to use the information lattice to find physical scramblers that generates algebraically correlated states. And then the bet is, the Fisher information of Arnau would be super efficient in picking up the criticality of the scrambling-purifying transition in these monitored quantum circuits.

NOTE 2 (Food for further thoughts:). *Before moving on to the details, here is a few pointers, that I personally think are really interesting food for thought for the whole game of measurement induced phase transitions in general. Although, obviously the finite size scaling of entanglement entropy growth seems to show a criticality, very clearly this is not at the level of states, as absolutely no states are conformal at any point in the phase diagram. That raises a few questions, for example, is this actually even second order, and if it is, then what is the relevant unconstrained phase-space. This concerns are of course not new, as we know that this is also exactly why this kind of criticalities suffer from post-selection problems. It is also extremely related to the questions why constructing the replica theory for these systems needs a different normalisation (the Born Rule factor) and also why the resultant Landau-Ginzburg theories*

[Adam2021MeasurementEntanglementPRXQuantumTheTheoryOfEverythingPaper] *actually do not have the correct symmetries of the original problem* ¹.

2. Information Lattice

Building on the works in [3] The *Information Lattice* as a concept to systematically analyze the information content and entanglement structure of a quantum many-body state was first introduced by Artiago et. al. in [2], and further developed as a tool for dynamics, characterisation of localisation, rate of information scrambling, criticality of the localisation-scrambling transition, and quench dynamics respectively in [2, 4–6]. Here for the sake of completeness, we quote the construction of information lattice and some properties possibly useful to us from the aforementioned works.

3. Insights from Information Lattice calculations

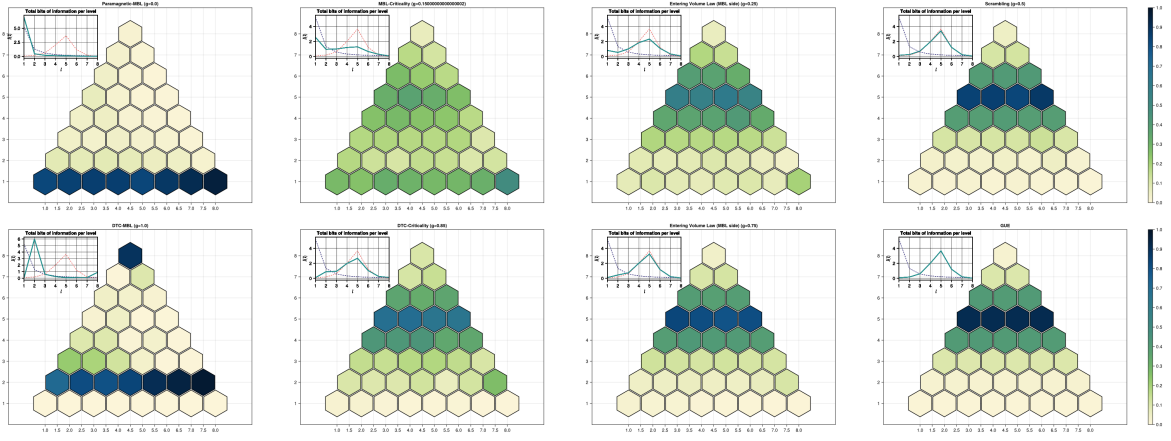


FIGURE 1. Information Lattice diagram for the time-crystalline circuit: showing the mutual information values for different parameter combinations. The color intensity represents the magnitude of the mutual information, with darker shades indicating higher values. This diagram helps visualize how information is distributed across various length scales of the system.

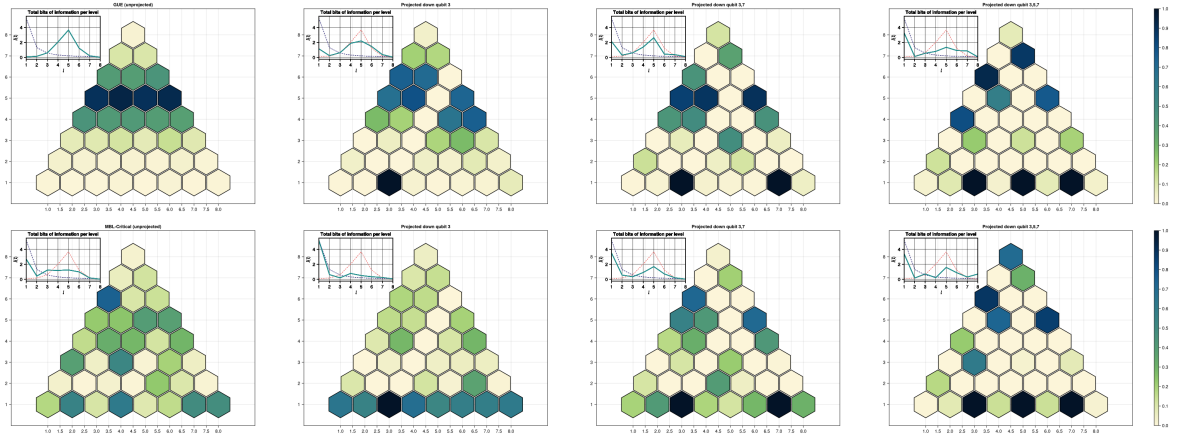


FIGURE 2. Information Lattice diagram for the ETH-MBL Critical state of the time crystalline circuit (top panel) and GUE (bottom panel) under **a**: no projections ,**b**: projections at site 3, **c**: site 3,5,**d**: site 3,5,7.

¹Adam Nahum, private communication, (*Damn! I have been waiting to write this line! xD*)

References

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4. The Circuit: Vanilla model of a Discrete Time Crystal

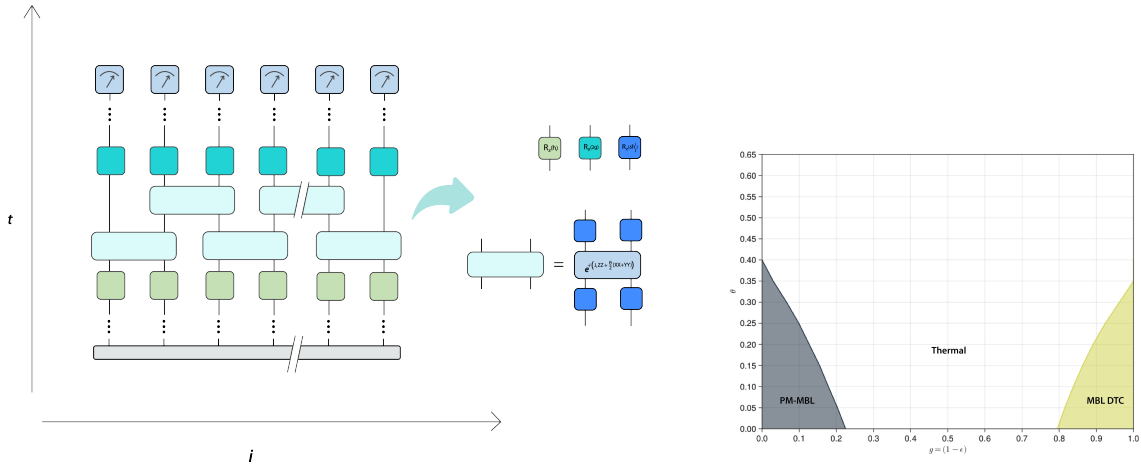


FIGURE 3. Schematic representation of the circuit and the associated phase diagram. The circuit exhibit three phases. For smaller values of θ and ϵ close to 1, that is g close to 0, it can support a MBL phase which is paramagnetic in nature. For small θ and ϵ close to 0, that is g close to 1, it supports a DTC MBL phase with a period multiplicity of 2. In the middle region the system tend to scramble. The phase boundary was extracted from results of [7], and was obtained from finite size scaling of the disorder averaged level spacing ratio.

The model we will be concerned with is the brick-wall circuit from Ipolitti et al [7, 8]. The General Floquet Model considered here, that exhibits a Discrete Time Crystal in 1D is an disordered Ising model with periodic $\pi/2$ Kicks about the x -axis. The system is probed only at stroboscopic times, $t = n\tau$, and $\tau = 1$

$$(1) \quad U_F = e^{-ig\pi \sum_i X_i} e^{-i\tau(\sum_i J_i Z_i Z_{i+1} + H_{int})} e^{-i\tau(\sum_i h_i Z_i)}$$

Here we have adopted the standard information theoretic notation (i.e have denoted Pauli operator $\sigma_i^Z = Z_i$ etc.) The H_{int} denotes some generic interaction such as fields or coupling,

$$H_{int} = \frac{\theta}{2} \sum_i [X_i X_{i+1} + Y_i Y_{i+1}] (\text{XY Coupling})$$

Among these parameters J_i, h_i are disordered and are uniformly sampled from the following intervals $J_i \in [0, \pi], h_i \in [0, 2\pi]$. The disorder is utilised to ensure localisation and thereby to prevent the system from heating up to triviality from the kicks [9]. Note that unlike the paramagnetic case, to stabilise time-crystalline MBL it is essential to have a disorder realisation on the Ising even couplings J_i , as otherwise the kicks cancels out the local background field over even periods (especially for small ϵ , and exactly at the perfect time-crystalline point) [7]. The parameter $g = (1 - \epsilon)$ controls the angle of the kick and forms one of the tuning parameters.

We also add further small uncertainties in the realisations of the two body gates

$$(2) \quad \theta = [\bar{\theta} - \frac{\Delta\theta}{2}, \bar{\theta} + \frac{\Delta\theta}{2}]$$

where $\Delta\theta = \frac{\pi}{50}$ and random independently drawn small single body rotation noise to each Z legs j of each gates i , $RZ(\delta h_j)$, where $\delta h_j \in \mathcal{N}(0, \frac{\pi}{50})$.

In absence of interaction and with perfect kicks it is trivial to see that operators of the form $\langle Z(0), Z(n) \rangle$ breaks the discrete time symmetry and exhibits a period doubling. The non-triviality of the DTC as a *phase* comes from the fact that this feature persists strongly even in the presence of interactions as well as with imperfect kicks (i.e $g = \frac{1}{2} = (\pi - \epsilon)$ with finite but small ϵ). In a quantum device this is realised as a brick-wall circuit. Below we quote two images from [8] that simulated portions of this circuits on a 20 qubit 1-D subsystem of the Google Sycamore processor. Other later experiments include [10] on 57 qubits of IBM Heron processor and exhibits similar results.

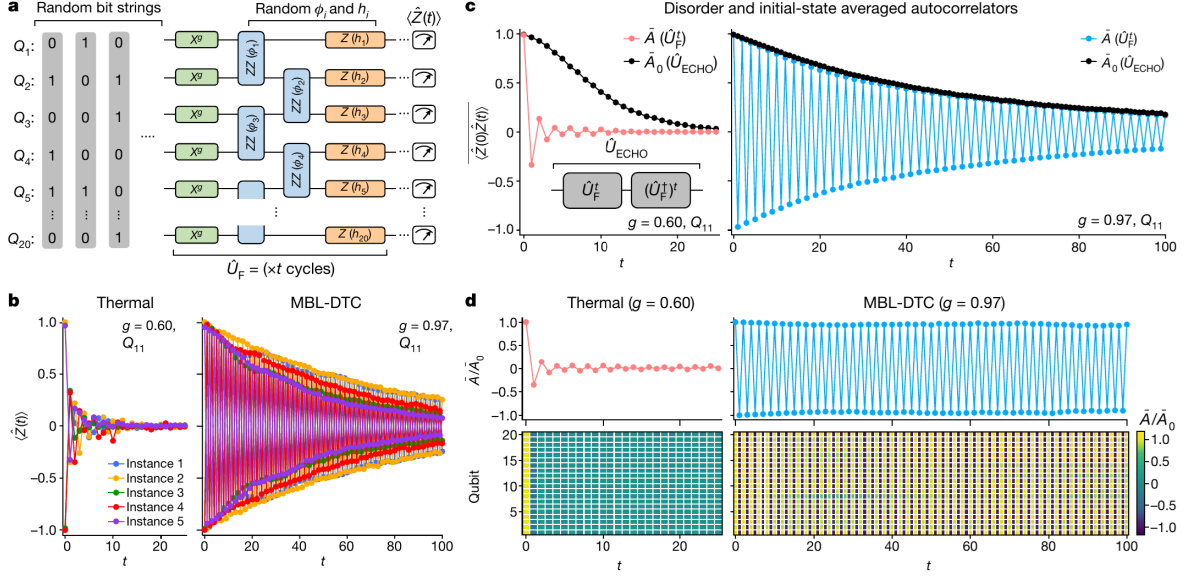


FIGURE 4. **Observing an MBL-DTC.** (a) The experimental circuit composed of t identical cycles of the unitary $\hat{U}_F$. The local polarization of each qubit, $\langle \hat{Z}(t) \rangle$, is measured at the end. In the following panels, we investigate a number of disorder instances each with a different random bit-string initial state. (b) Experimental values of $\langle \hat{Z}(t) \rangle$ measured at Q_{11} . Data are shown for five representative circuit instances deep in the thermal ($g = 0.60$; left) and MBL-DTC ($g = 0.97$; right) phases. (c) Autocorrelator $\bar{A} = \langle \hat{Z}(0)\hat{Z}(t) \rangle$ at Q_{11} , obtained from averaging the results of 36 circuit instances. For the same circuit instances, the average autocorrelator at the output of $\hat{U}_{\text{ECHO}} = (\hat{U}_F^\dagger)^t \hat{U}_F^t$ is also measured and its square root, $\bar{A}_0$, is shown alongside $\bar{A}$ for comparison. The left (right) panels correspond to $g = 0.60$ (0.97). (d) Top panels: the ratio $\bar{A}/\bar{A}_0$ obtained from (c). Bottom panels: $\bar{A}/\bar{A}_0$ as a function of t and qubit location. The left (right) panels correspond to $g = 0.60$ (0.97).

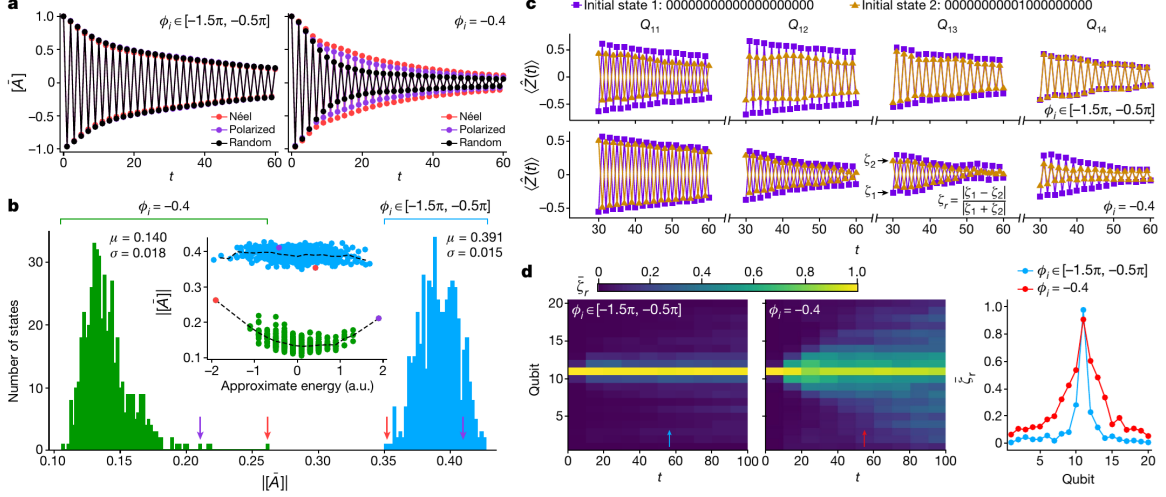


FIGURE 5. **Distinguishing MBL-DTC from prethermal phenomena.** (a) Site- and disorder-averaged autocorrelators $\bar{A}$ measured with $g = 0.94$. In the left panel (MBL-DTC), each dataset is averaged over 24 disorder instances of ϕ_i and h_i , with the initial state fixed at one of the following: Néel, $|01\rangle^{\otimes 10}$; polarized, $|0\rangle^{\otimes 20}$; random, $|00111000010011001111\rangle$. In the right panel (prethermal), the same values of h_i and initial states are used but $g_i = -0.4$. (b) Histograms of $|\bar{A}|$, from 500 random bit-string initial states, averaged over cycles 30 and 31 and the same disorder instances as in (a). The standard deviation (mean) of $|\bar{A}|$, σ (μ), is also listed. Location of the polarized (Néel) state is indicated by a purple (red) arrow. *Inset:* same collection of $|\bar{A}|$ plotted over the energies of the bit-string states, calculated from the effective Hamiltonian $\hat{H}_{\text{eff}}$ approximating the drive (see text). Dashed lines show averaged values within energy windows separated by 0.2. (c) $\langle \hat{Z}(t) \rangle$ for two bit-string initial states that differ only at Q_{11} . Top panel shows a single circuit instance with disordered ϕ_i , and bottom panel shows an instance with uniform $\phi_i = -0.4$. (d) Left and middle panels: relative difference between the two signals ζ_r as a function of t and qubit location, averaged over time windows of 10 cycles and over 64 disorder instances for $\hat{U}_F$ and 81 instances for $\hat{U}_F$. Right panel: qubit dependence of ζ_r , averaged from $t = 51$ to $t = 60$.